

Title:

Continuous Propagation of Ground Motion Model Uncertainty in Probabilistic Seismic Hazard Analysis

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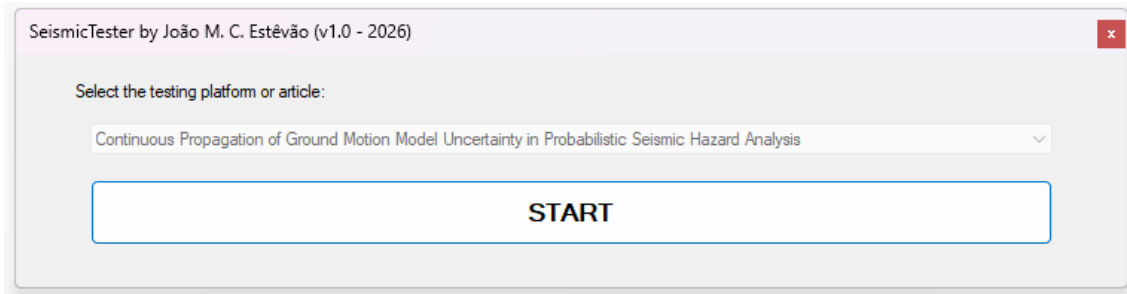
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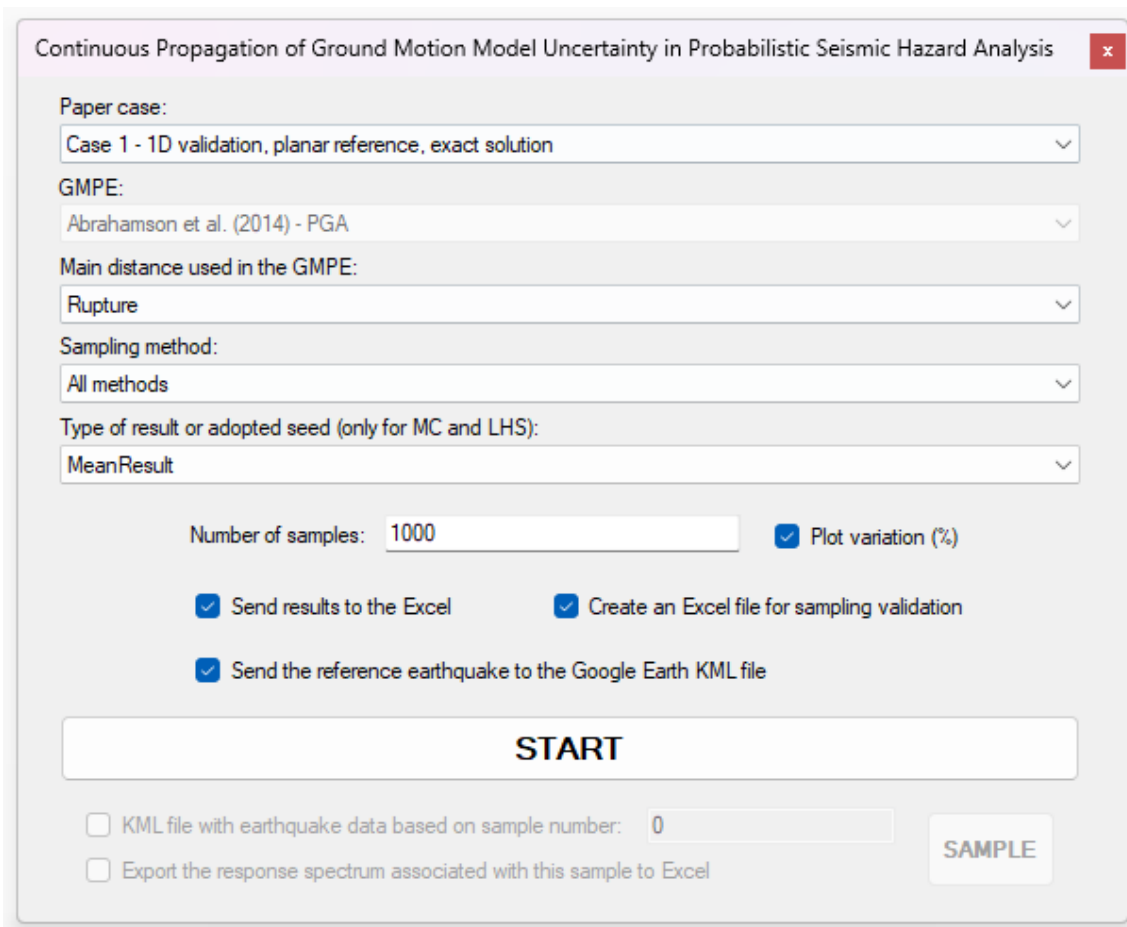
Application interface overview

The compiled application provides a direct and controlled interface to the computational framework described in the paper.

The initial screen fixes the configuration to the present study, ensuring that all subsequent operations are fully consistent with the methodologies and assumptions used to generate the published results.



After initialization, the main interface allows the user to select the case study, define sampling strategies, and configure key parameters governing the propagation of GMPE uncertainty.



All numerical experiments are executed through this interface, and the resulting outputs (plots, Excel files, and KML visualizations) are generated automatically, enabling full reproducibility and inspection of the computational process.

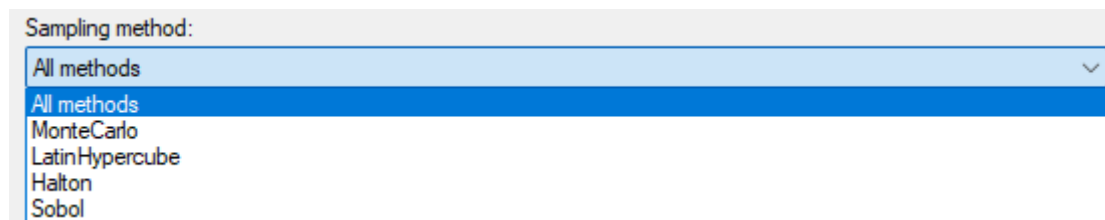
This design ensures that all numerical results can be reproduced, inspected, and validated independently.

Sampling strategies

The application provides explicit control over the sampling strategy used to propagate internal GMPE uncertainty, including Monte Carlo simulation, Latin Hypercube Sampling, and low-discrepancy deterministic sequences (Halton and Sobol).

The option **All methods** enables direct comparison under identical configurations, allowing the user to assess the impact of the sampling procedure on the numerical results.

This functionality reflects the central aspect of the proposed framework, where variability introduced by the sampling process must be clearly distinguished from the underlying physical model uncertainty.



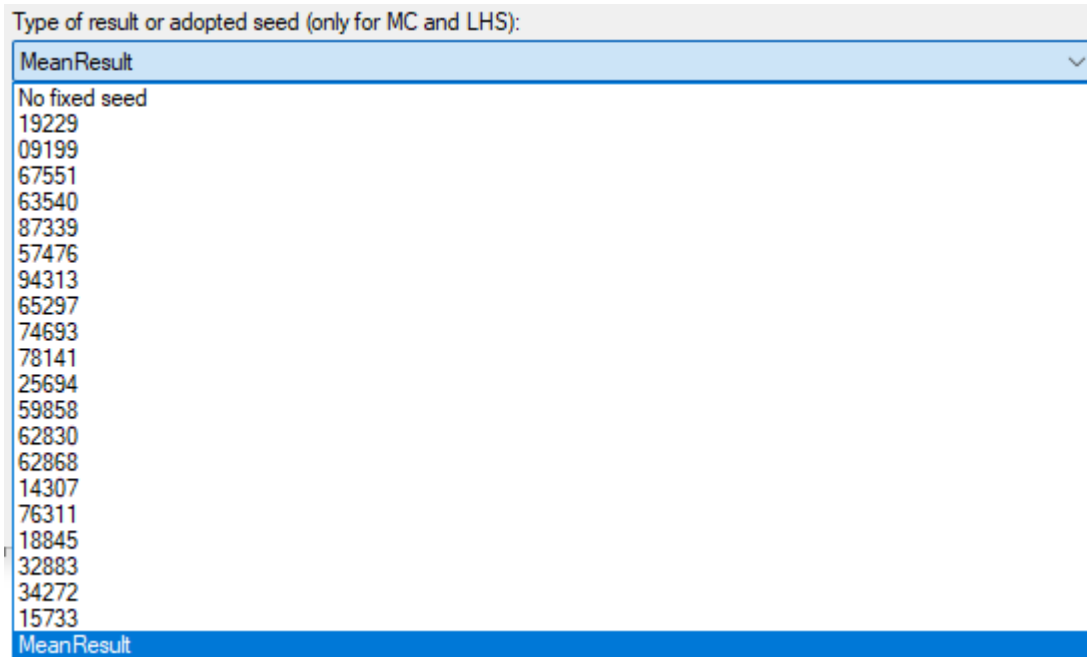
Control of sampling variability

For stochastic sampling methods (Monte Carlo and Latin Hypercube), the application provides explicit control over the random seed and aggregation of results.

Individual seeds allow the user to reproduce specific realizations of the sampling process, making it possible to analyze variability associated with finite samples. The option **No fixed seed** enables unrestricted stochastic sampling.

The option **MeanResult** provides an aggregated estimate that reduces sampling-induced variability, approximating the expected value of the computed quantities.

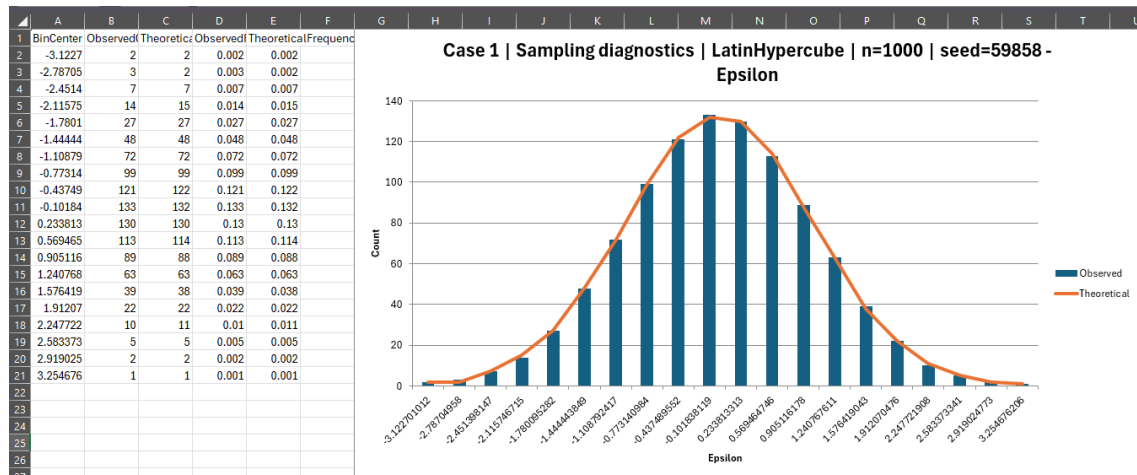
This functionality makes it possible to explicitly distinguish variability introduced by the sampling procedure from the underlying physical model uncertainty, which is a central aspect of the proposed framework.



Sampling diagnostics and validation

The application provides explicit diagnostics of the sampling process, allowing direct comparison between the observed distribution of sampled variables and their theoretical counterparts.

The figure illustrates a representative example for the internal variable ε (standard normal distribution), using Latin Hypercube Sampling with $n = 1000$ samples.



The close agreement between observed and theoretical distributions confirms the correct implementation of the sampling procedure and highlights the ability of the framework to distinguish numerical variability from the underlying probabilistic model.

Such diagnostics are automatically generated for all sampling strategies, enabling validation of the numerical process and full transparency in the propagation of GMPE uncertainty.

Physical interpretation and sample-level inspection

The application enables direct inspection of individual samples through geospatial visualization.

Each sampled configuration can be exported to a KML file and visualized in Google Earth, where all geometric and distance parameters associated with the rupture are explicitly represented. By interacting with the graphical elements, the user can access the numerical values used in the GMPE evaluation, such as rupture-to-site distances.

☒ KML file with earthquake data based on sample number:

☒ Export the response spectrum associated with this sample to Excel

SAMPLE

This functionality establishes a direct link between the abstract probabilistic sampling space and physically interpretable configurations, allowing detailed inspection of individual realizations.

It further enables full traceability and independent verification of the numerical results at the level of individual samples, complementing the statistical diagnostics of the sampling process.

